

D623

Public Health Finance and Funding

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A federal agency awards a state \$2 million to build an overdose surveillance system. The agency will approve key staff, co-develop the data plan, and join monthly planning calls. The award letter arrived on the state's fiscal year-end. What type of award is this?

- A. Contract.
- B. A philanthropic gift.
- C. Cooperative agreement.
- D. Grant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A health department's finance director, who has 20 years of grants experience, prepares a proposal with \$400,000 in direct costs. The negotiated indirect rate is 20% of direct costs. What is the total request?

- A. \$400,000
- B. \$480,000
- C. \$500,000
- D. \$80,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A grant is capped at \$600,000 total, including indirect costs at 20% of direct costs. What is the maximum for direct costs?

- A. \$720,000
- B. \$480,000
- C. \$120,000
- D. \$500,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A health educator with a master's degree and 10 years of experience earns \$70,000 a year and spends 40% of her time on a diabetes grant. How much of her salary may be charged to the grant?

- A. \$28,000
- B. Any amount, because her experience makes it reasonable.
- C. \$70,000
- D. \$42,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A program requires a local match of 20% of total project costs. The federal share is \$400,000. The county plans to count donated space in a church hall toward match. What match amount is required?

- A. \$100,000
- B. \$500,000
- C. It cannot be met, since donated space never counts.
- D. \$80,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A county's \$60,000 match exactly meets a requirement of 25% of total project costs. What is the federal share?

- A. \$45,000
- B. \$15,000
- C. \$240,000
- D. \$180,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A county has paid two school nurses from local taxes for eight years. It wins a new grant and plans to move both nurses onto the grant and cut the local nursing budget. The grant was featured in the local paper. Is this acceptable?

- A. Yes, since the public already knows about the grant.
- B. No; it supplants local funds instead of adding capacity.
- C. Yes, grants may always pay existing staff.
- D. Yes, because the nurses serve the same purpose.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A program that won a state award this year budgeted \$200,000 for personnel and spent \$185,000. It budgeted \$30,000 for supplies and spent \$36,000. What is the supplies variance?

- A. \$6,000 unfavorable (16.7% over budget).
- B. \$6,000 unfavorable (20% over budget).
- C. \$15,000 favorable.
- D. \$6,000 favorable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A travel line came in 10% under budget, with actual spending of \$45,000. What was the budget?

- A. \$50,000
- B. \$49,500
- C. \$45,000
- D. \$40,500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A reviewer flags a budget line reading only “Supplies: \$12,000.” The applicant's prior grant received top marks. What revision best fixes the line?

- A. Raise it to \$15,000 for safety.
- B. Note that the prior grant was rated highly.
- C. Lump supplies into “Other.”
- D. Itemize, e.g., 400 glucose test kits × \$30 for screening events.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Program A costs \$200,000 and prevents 40 asthma hospitalizations. Program B, which uses a mobile app built by a local startup, costs \$380,000 and prevents 70. What is the incremental cost-effectiveness ratio of B compared with A?

- A. \$5,429 per hospitalization prevented.
- B. \$180,000.
- C. \$6,000 per additional hospitalization prevented.
- D. \$0.00017 per hospitalization.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A new program adds \$90,000 in cost over the current one, with an ICER of \$3,000 per extra case prevented. How many extra cases does it prevent?

- A. 3,000
- B. 0.033
- C. 30
- D. 93,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A federal agency awards a state \$2 million to build an overdose surveillance system. The agency will approve key staff, co-develop the data plan, and join monthly planning calls. ~~The award letter arrived on the state's fiscal year-end.~~ What type of award is this?

Target word: award type **Decides:** Substantial funder involvement.

A	Confusion set	A contract buys a product for the agency itself.
B	Inverted operation	The funder is a federal agency.
C	Correct	Substantial federal involvement defines it.
D	Adjacent category	A grant has little funder involvement.

2. Answer B Concept 2

A health department's finance director, ~~who has 20 years of grants experience,~~ prepares a proposal with \$400,000 in direct costs. The negotiated indirect rate is 20% of direct costs. What is the total request?

Target word: total request **Decides:** $400,000 + 20\% \times 400,000$.

A	Dropped element	Leaves out indirect costs.
B	Correct	$400,000 + \$80,000 = \$480,000$.
C	Inverted operation	Divides by 0.8 instead of adding 20%.
D	Confusion set	Only the indirect portion.

3. Answer D Concept 2 · transfer

A grant is capped at \$600,000 total, including indirect costs at 20% of direct costs. What is the maximum for direct costs?

Target word: solve for direct **Decides:** $\text{Direct} \times 1.2 = 600,000$.

A	Inverted operation	Multiplies instead of dividing.
B	Adjacent category	Takes 20% off the total instead of dividing by 1.2.
C	Wrong stem number	20% of the total, not the direct base.
D	Correct	$\$600,000 \div 1.2 = \$500,000$.

4. Answer A Concept 3

A health educator ~~with a master's degree and 10 years of experience~~ earns \$70,000 a year and spends 40% of her time on a diabetes grant. How much of her salary may be charged to the grant?

Target word: allocable salary **Decides:** $40\% \times \$70,000$.

A	Correct	$\$70,000 \times 0.40 = \$28,000$.
B	Camouflage	Experience does not change allocation.
C	Dropped element	Charges full salary for part-time work.
D	Inverted operation	Uses the 60% not spent on the grant.

5. Answer A Concept 4

A program requires a local match of 20% of total project costs. The federal share is \$400,000. ~~The county plans to count donated space in a church hall toward match.~~ What match amount is required?

Target word: required match **Decides:** Total = $400,000 \div 0.8 = 500,000$.

A	Correct	$\$500,000 - \$400,000 = \$100,000$.
B	Confusion set	That is the total project cost.
C	Absolute language	In-kind contributions often count as match.
D	Adjacent category	20% of the federal share, not of the total.

6. Answer D Concept 4 • transfer

A county's \$60,000 match exactly meets a requirement of 25% of total project costs. What is the federal share?

Target word: solve for federal share **Decides:** Total = $60,000 \div 0.25 = 240,000$.

A	Confusion set	75% of the match.
B	Inverted operation	Multiplies the match by 25%.
C	Dropped element	The total, not the federal share.
D	Correct	$\$240,000 - \$60,000$.

7. Answer B Concept 5

A county has paid two school nurses from local taxes for eight years. It wins a new grant and plans to move both nurses onto the grant and cut the local nursing budget. ~~The grant was featured in the local paper.~~ Is this acceptable?

Target word: supplanting **Decides:** Grant replaces existing local funding.

A	Camouflage	Publicity does not affect the rule.
B	Correct	Grants must supplement, not replace.
C	Absolute language	Only when it adds effort without cutting local support.
D	Adjacent category	Purpose alignment does not permit supplanting.

8. Answer B Concept 6

A program ~~that won a state award this year~~ budgeted \$200,000 for personnel and spent \$185,000. It budgeted \$30,000 for supplies and spent \$36,000. What is the supplies variance?

Target word: supplies variance **Decides:** $30,000 - 36,000 = -6,000$; $6,000 \div 30,000$.

A	Confusion set	Divides by actual instead of budget.
B	Correct	Overspent by \$6,000, 20% of the budget.
C	Wrong stem number	That is the personnel variance.
D	Inverted operation	Overspending a cost line is unfavorable.

9. Answer A Concept 6 · transfer

A travel line came in 10% under budget, with actual spending of \$45,000. What was the budget?

Target word: solve for budget **Decides:** Actual = 0.9 × budget.

A	Correct	$\$45,000 \div 0.9 = \$50,000$.
B	Adjacent category	Adds 10% of actual instead of dividing by 0.9.
C	Dropped element	Ignores the variance.
D	Inverted operation	Takes 10% off the actual.

10. Answer D Concept 7

A reviewer flags a budget line reading only “Supplies: \$12,000.” ~~The applicant's prior grant received top marks.~~ What revision best fixes the line?

Target word: justification fix **Decides:** Needs computation and link to activity.

A	Inverted operation	Padding without explanation weakens it.
B	Camouflage	Past success does not justify this line.
C	Dropped element	Removes detail instead of adding it.
D	Correct	Shows the math and purpose.

11. Answer C Concept 8

Program A costs \$200,000 and prevents 40 asthma hospitalizations. Program B, ~~which uses a mobile app built by a local startup,~~ costs \$380,000 and prevents 70. What is the incremental cost-effectiveness ratio of B compared with A?

Target word: ICER **Decides:** $(\$380,000 - \$200,000) \div (70 - 40)$.

A	Adjacent category	Program B's average cost, not the incremental ratio.
B	Dropped element	Extra cost without dividing by extra effect.
C	Correct	$\$180,000 \div 30 = \$6,000$.
D	Inverted operation	Divides effect by cost.

12. Answer C Concept 8 · transfer

A new program adds \$90,000 in cost over the current one, with an ICER of \$3,000 per extra case prevented. How many extra cases does it prevent?

Target word: solve for effect **Decides:** $90,000 \div 3,000$.

A	Wrong stem number	The ICER itself.
B	Inverted operation	Divides the ICER by the cost.
C	Correct	$\$90,000 \div \$3,000 = 30$.
D	Confusion set	Adds the two numbers.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Grants, cooperative agreements, and contracts	A grant funds the recipient's project with little funder involvement; a cooperative agreement adds substantial funder involvement in the work; a contract buys specific goods or services for the funder.	Treating any federal award with reporting rules as a cooperative agreement.	How much the funding agency will take part in running the project.
2	Direct and indirect costs	Indirect costs = rate × the direct cost base; total request = direct + indirect.	Forgetting to add indirect costs, or backing out the direct base by multiplying instead of dividing.	A negotiated indirect rate and a direct cost figure or a total cap.
3	Allowable, allocable, reasonable	A cost may be charged to a grant only if it is allowed by the rules, reasonable in amount, and allocated in proportion to the benefit the grant receives.	Charging a full salary for someone who works part-time on the grant.	A staff member's percent effort on the project.
4	Cost sharing and match	When match is a percent of total project cost, total = federal share ÷ (1 – match rate), and match = total – federal share.	Applying the match percent to the federal share instead of the total.	“Of total project costs” in the match rule.
5	Supplement, not supplant	Grant funds must add new capacity, not replace local funds already paying for the same work.	Using a grant to pay existing staff so local dollars can be cut.	Grant money going to positions or services that were already locally funded.
6	Budget variance analysis	Variance = budget – actual for a cost line; spending over budget is unfavorable; percent variance is divided by the budgeted amount.	Calling overspending favorable, or dividing by actual.	Budgeted and actual amounts for a line item.
7	Budget justification	Each budget line needs a computation basis (quantity × unit cost) and a direct link to a planned activity.	Listing a lump sum with no calculation or purpose.	A reviewer comment on a single-number line item.
8	Cost-effectiveness (ICER)	Incremental cost-effectiveness ratio = (cost of new – cost of current) ÷ (effect of new – effect of current).	Using one program's average cost per outcome instead of the incremental ratio.	Two programs with costs and outcomes compared.

Course scope (public description): Public Health Finance and Funding offers students a field-specific approach to acquiring the budgetary skills needed to secure funding for public health programs and manage it in prudent ways. Students learn to navigate various funding sources, such as government grants and cooperative agreements, and philanthropic organizations. Students learn to create budgets using government documentation both for managing funds and for persuading funding organizations of the value of projects. Finally, students conduct a variance analysis—the process by which budgets are evaluated for feasibility after programs have launched. There are no prerequisites for this course.